Q1.

Diagram 1 below shows the lungs and the trachea, the airway leading to the lungs. One of the lungs is drawn in section.

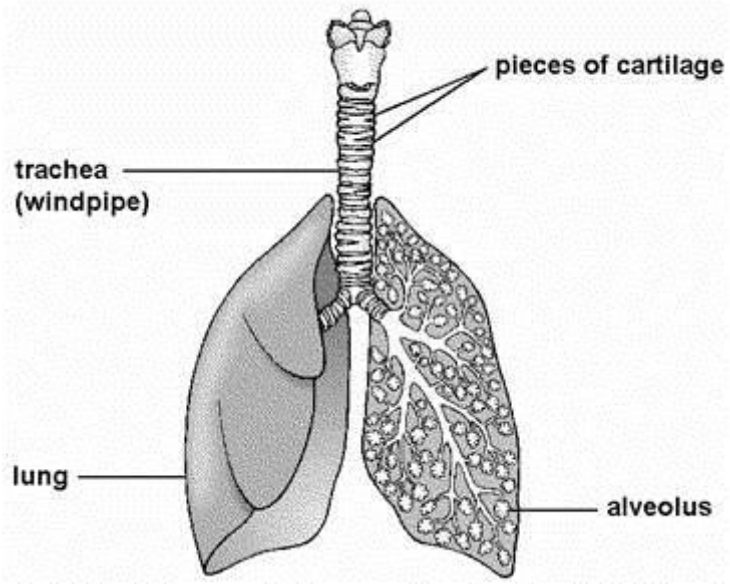


diagram 1

(a) In the wall of the trachea, there are pieces of a stiff material called cartilage.

Why is this stiff material necessary in the wall of the trachea?

.....

.....

1 mark

(b) Diagram 2 below shows one alveolus and its blood supply.

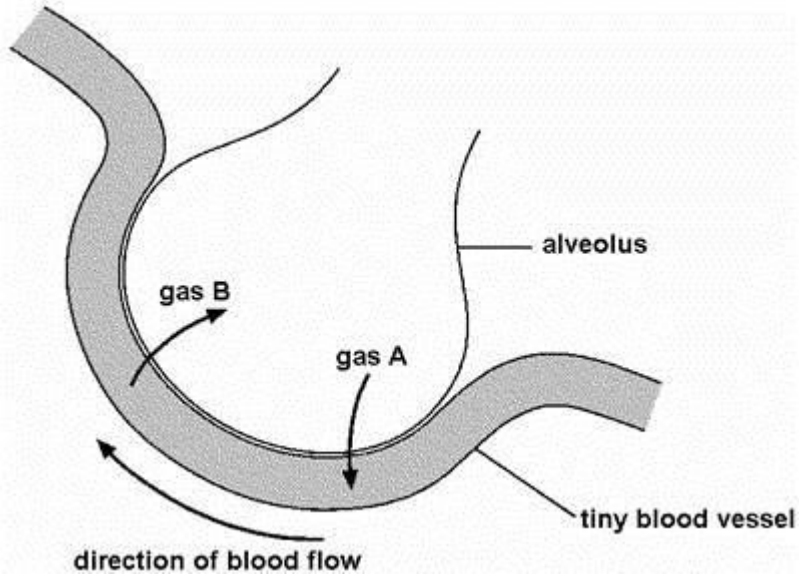


diagram 2

(i) Look at diagram 2, above.
Gas A **enters** the blood from the alveolus.
Gas B **leaves** the blood and enters the alveolus.
What are the names of gases A and B?

gas A .....

gas B .....

1 mark

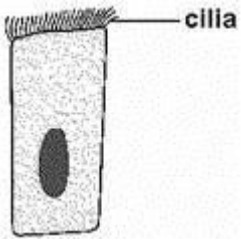
(ii) Give **one** reason why it is easy for gases to pass across the wall of an alveolus.

.....

.....

1 mark

(c) The diagram below shows a ciliated cell from the lining of the airway.



(i) What is the function of this cell in the airway?

.....

.....

1 mark

(ii) This cell is affected by substances in cigarette smoke.
What effect does cigarette smoke have on the cilia?

.....

.....

1 mark

(iii) Give the name of the substance, in cigarette smoke, which causes addiction to smoking.

.....

1 mark
Maximum 6 marks

Q1.

(a) any **one** from

- to prevent it collapsing
  accept 'protects against collapse'
- to keep it open
  'for protection' is insufficient
- for support
  accept 'for strength'
  accept 'for flexibility'

1 (L5)

(b) (i) *A*: oxygen
accept 'O_2'

B: carbon dioxide
accept 'CO_2'
***both** answers are required for the mark*

1 (L5)

(ii) any **one** from

- it is thin
- it is one cell thick
- it is close to the blood supply
  accept 'there is a diffusion gradient'
  accept 'it is moist'

1 (L6)

(c) (i) any **one** from

- it moves mucus
  accept 'it moves bacteria'
- it sweeps dust from lungs
  'to clear or clean the airways' is insufficient

1 (L6)

(ii) any **one** from

- it paralyses the cilia
- it stops the cilia working
- it clogs the cilia
  accept 'it destroys them'
  *do **not** accept 'it kills cilia'*

1 (L5)

(iii) nicotine

1 (L6)

[6]